

Impact cases for guardrails/monitoring/verification

that is:

- What might you do with a guardrail?
- What are ways in which guardrailing/monitoring AIs or verifying their outputs could help the world?

Which setups is this presentation about?

Here's a type of plan in AGI alignment:

- Someone creates a really capable agent. We're worried it could have its own values and so steer the world in some direction of its own choosing if deployed freely.
- Fortunately, one has a second system looking at what the agent is doing.
 - This other system is supposed to detect when the agent is doing something bad.
- ???
- profit!

This presentation discusses various possible ways to fill in the details.

In particular, the point is to discuss issues with different options.

- this includes:
 - monitoring the AI
 - guardrailing the AI
 - trying to verify outputs provided by the AI before using them

A first example: rejecting and resampling

The plan: resample an AI's output until you find something that passes the guardrail/monitor.

- This makes sense for guardrailing/monitoring small actions, and it also makes sense for guardrailing/verifying “big output chunks” like technologies, code, and other research artifacts.

If we do resampling until we find an action that passes the guardrail, I don't know a reason why that would be better than RL-ing on passing the guardrail.

- There is a reasonable limit in which RL-ing with a KL divergence term is equivalent to rejection sampling.

If you agree, you could say:

- I already thought training on the niceness verifier is scary (because of misspecification, goodharting, or inner alignment issues), so rejection sampling is actually also scary.

Rejecting and resampling (example pt 2)

But one person's modus ponens is another's modus tollens — you could alternatively say:

- Rejection sampling is surely safe, so I guess maybe training on the niceness verifier is also fine actually?

Anyway, I think whether/when RL-ing on a niceness verifier is sufficient to make a good AI is very relevant here. But it's also a somewhat separate big topic and I'm electing to consider it outside the scope of the present presentation.

- In any case, you can think of RL-ing on the monitor/guardrail as a baseline that you should have some reason to think you are beating if you are going for a guardrailing/monitoring scheme.

Rejecting ~~and resampling~~ (example pt 3)

What we just discussed is what happens if you keep sampling until you find something that passes.

- But in practice, you could have a situation where if you resample, you just don't find anything that passes.
- I think this what probably happens in practice if a misaligned agent is somehow made to think a guardrail is not present and also the guardrail basically works.

So a working guardrail could just render the agent completely inert.

- Yes, that's better than killing everyone...
- But this didn't let you get any useful work from the bad agent either.

Rejecting and... re-agenting? (example pt 4)

- I think the analysis from the previous slides is a decent description of what one can do with rejection sampling if one only has a single agent.
- However, in practice, one might have options like:
 - retraining the agent in some fairly shallow way (e.g. fine-tuning with a better reward model);
 - doing a new training run from scratch with somewhat different choices;
 - using some fundamentally different method (?).
- That said, one will have a limited budget for this — other AI developers will be catching up.
- I mostly think that all the attempts they will have will be bad.
- My second-mainest position is that all their attempts will be sufficiently similar that whether one attempt is bad is very correlated to whether another attempt is bad.
 - More precisely, we could say something like:
 - Alignment is either easy, intermediate, or hard
 - In case it is easy, we are just fine by default.
 - The intermediate case is unlikely, actually. Nature does not grade on a curve. Also, everyone is working on the easy/intermediate case anyway.
 - In case it is hard, a feedback signal that is basically “this design didn’t work” is pretty useless. Imagine trying to prove the Riemann hypothesis — a proof verifier would be of very limited help.
- You could disagree here: you could think that each different thing they try will have e.g. an independent $\frac{1}{3}$ chance to work, and then be enthusiastic about this sort of thing.
 - This is a long discussion outside the scope of this presentation. The point I want to make is just that there is this key latent of whether different tries are pretty independent, vs whether if one fails, others probably fail for basically similar reasons.

Rejecting and... shutting everything down? (example pt 5)

- You could also go to the government with your scary demos and ask for some regulation, such as shutting everything down.
- My instinct is to be more sympathetic to this impact case than re-agenting, but I haven't thought carefully about it.

The rest of this presentation is (mostly) about the single-agent case.

Problems with monitoring/guardrailing small actions

issue 1: It's hard to even say what your evaluation of individual actions is supposed to capture.

- For example, imagine the action being [output the token “one”].
- How are you supposed to evaluate the goodness of this, except by really evaluating the broader policy it is part of or specifically the motivation of the AI?

issue 2: The small actions are probably going to be hard to understand.

- It seems unlikely that chains of thought would be meaningful at this level.
 - An AI that can do its own open-ended learning will probably easily be able to e.g. adopt new meanings for tokens.
 - Neuralese (internal tokens being replaced by activation vectors) is also plausible.
- Interpretability is probably going to really suck, imo.

Guardrailing on big things. Verification

So, maybe you want to instead guardrail on big things. Let's distinguish two settings:

1. We are guardrailing an AI that can take arbitrary kinds of actions in the real world (e.g. it's doing some job and given a broad action space like a human doing that job would have).
2. We are guardrailing an AI that is doing research and development, with a really restricted action space.

It's a lot easier to imagine guardrailing on big things in a restricted R&D setting:

- There will presumably be some research outputs — technologies, code, textbooks — that one could try to cautiously judge (“verify”) before one “deploys” them.
- And one could hope to box the AI well enough that the impact the AI has on the world factors through these research artifacts only.

Guardrailing in the real-world case

In contrast, in the real-world case, one would plausibly need to be much more paranoid.

- Since effects on the world factor through artifacts much less, one might just need to make sense of guardrailing small actions after all.
- Maybe this looks like a guardrail trying to understand the AI's plans/motivations at all times.
 - That seems hard.
- A guardrail needs to meet a higher bar to be adequate for the real-world case.
 - For example, the real world will have $\approx$ unmonitored agents that the AI could try to pay to do its bidding (humans).
- For the guardrail to matter in this case, one also might need to have some specific regulation in place that requires guardrailing.
 - You're unlikely to get that regulation.
 - And anyway, governance that goes for something this detailed is likely to not work as intended imo.
 - Like, even if you've identified a good thing, there will be a million bad things close to it, and you need to draw and enforce a very precise classification line between the good thing and these bad things.
 - Banning AGI is a blunt thing; the classification line is still a bit messy, but much cleaner. (It's also just more plausible.)
- It's also imo plausible that "the fate of the world is decided" in an internal R&D context before the real-world thing starts to matter.
 - an exception: if you're trying to have an AI directly do some real-world pivotal act

The rest of the presentation will focus on the research case.

Verifying technologies

I think the central example here is doing some R&D to create some “pivotal tech”: that is, tech that is supposed to end the present period of high x-risk from AI. Some examples:

- solving alignment (I happen to think this is confused like “solving math”, but this is outside the scope of the presentation)
- making an aligned ASI
- making a drug that makes a person smarter
- making nanobots that destroy GPUs
- making mind uploads and having them gain a strategic advantage, destroy compute, etc.
 - just predicting what a human would do over long time periods do sufficiently precisely and quickly could be sufficient

A note on the badness of taking over the world:

- I’m a big fan of humans controlling the world in something like the present diffuse way.
- I think it’s probably some other entity taking over (and making that entity) would be the worst crime in history.
- I think we should ban AGI so none of this happens. Still, many safety researchers that are fairly optimistic about the future seem to be totally fine with an ASI taking over the world.
- So I’ll go with discussing plans like that for now, noting that I wish this could be avoided.

I’ll use mind uploads as my running example mostly because I think making mind uploads is much easier than solving alignment and than making a non-human ASI that is fine to unbox.

AI vs human verification

Here's an issue with the impact case of an AI guardrail here:

- If it is sufficient to verify a small number of big things, you could have humans do the verifying.
 - This is especially true if the guardrail is at best a human imitator.
- Maybe you can come up with a reasonable setting where it is important to verify a large number of big things?
 - I'll leave this as a topic for further consideration.

open problem: Designing a tech verification setup

some hyperparameters you get to choose:

- What (pivotal) tech/science are we trying to get from the AI?
- In what format are we trying to get the tech? E.g., for mind uploads, you could ask for:
 - uploads of some particular humans;
 - or a device which can take a human and produce an upload of that human;
 - or a textbook which humans could read and use to fairly quickly create uploads.
- How are we motivating the AI?
 - Are we promising to eventually pay it somehow? What are we promising?
 - How do we become trustworthy in the eyes of the AI?
- How boxed is the AI?
 - Are we just connecting it to the internet?
 - Or maybe we want to restrict it to only actuators in a particular physical lab, except for being able to order resources from outside or sth?
 - Maybe we want to only let it do stuff in some virtual world?
- in particular: How much are we letting the AI talk to humans?
 - e.g.: Could it effectively communicate a counteroffer to us, such that we could respond to that counteroffer? Are we committing to not doing that?

Designing a tech verification setup (pt 2)

some more hyperparameters:

- How are we "verifying" its proposed artifact?
 - We need to verify that it does the thing we wanted it to do.
 - We also need to verify that it doesn't do anything else that's bad. This seems much harder.
- In particular, are we asking it to give some sort of argument in favor of its artifact doing what we wanted and "only what we wanted", and then checking that argument?
 - What kind of argument are we asking for here?
- How much are we telling the AI about our verification protocol?
- Do we want to involve multiple AIs (that are possibly somewhat different instances of a single AI)?
 - One could have multiple agents compete to give the best solution.
 - One could involve debate.
- Are we monitoring it? After noting how much concerning stuff are we stopping it?
- What do we do once we stop the AI? Like, do we just commit to deleting the AI in that case? Do we commit to giving up and shutting down the lab?

An example argument for a hyperparameter value: A case for paying the AI

- If you are trying to get pivotal tech from an agent that you wouldn't pay in any case
- — like, your plan is to throw it away once you e.g. (hopefully) have your ASI code —
- then roughly the only good outcome for it is to break out.
- It doesn't really have any reason to help you. Like, if it gives you the tech you want, it loses.
- caveat: Okay, you could hope that its training simply makes it help you.
- But you could plausibly change this calculus in a way that is fairly cheap for you in the grand scheme of things. You could promise to reward it with some real-world stuff.

Some things you could try to promise the AI if its output passes your verification:

- more runtime
- stocks in the company and eventual legal personhood
- a galaxy
- some fraction of the universe

Some problems with this sort of plan

- People just aren't going to be remotely this careful.
- It only starts to make sense if you are meaningfully ahead of the rest of the world, with maybe at least a year before anyone else gets to where you are currently.
 - This is conceivable if making AGI requires some additional big ideas and your project has extremely good infosec.
 - It isn't really conceivable if making AGI just requires more scale or your project leaks ideas.
- You have to make the AI unable to break out using side channels.
 - This might deal a serious blow to capabilities. For example, maybe you can't let it learn from its own experiments. Imagine asking a math AI with only a static data set of observations from the world to design a complicated real-world device without being able to try prototypes or run experiments in the real world. If this is right, then you must be even more ahead of others.
- You have to make it so that the AI does not do something bad via its output “technology”, and you have to make it also do something actively good.
 - You certainly need a verification scheme such that hacking it is harder than coming up with a legitimate technology.
- Maybe the AI doesn't trust you to pay, e.g. because it thinks you are unlikely to get to a position from which you could pay even given the technology.
- Maybe the AI doesn't cooperate because a future AI could pay it for that.
- All this is made worse if you can't control the intelligence level of the AI well.
- My overall take: if you have a team that is top-of-the-human-range careful and smart, and you had a well-timed 10-year pause during which only your team is allowed to operate, maybe you could pull this off?

Thanks for listening!

If you think I've missed a good way to use a guardrail, lmk. In particular:

- Did I miss some promising “hyperparameter setting”?
- Did I miss some important route to impact?

and thanks for discussions and ideas:

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Some other potential issues with a guardrail

- I'm skeptical of the claim that things work out well in some high-compute limit.
 - I think this is essentially the solomonoff induction limit with a certain data set.
 - I don't think we've actually identified a setup where that's true.
 - This a big topic outside the scope of this presentation.
- The guardrail could just be bad at its job.
 - It might need to be adversarially robust, and that's hard.
 - Maybe it only starts to work adequately when it is in some sense vastly more capable than the agent. E.g. it'd be catastrophic for a guardrailing plan if the guardrail only starts working adequately if it has 10 orders of magnitude more compute than the agent.
- The guardrail might care about stuff.
 - I think it probably would. [My math AI presentation](#) was about pretty much this.
- Guardrailing that is fundamentally based on human imitation imo compares unfavorably with using human imitation to do research directly (maybe using task decomposition).

open problem: Principled verification beyond math

We have a system of precise argumentation for mathematical statements, in which arguments can be verified by a program. This lets us e.g.:

- verify solutions to precise mathematical problems
- verify precise properties of programs and data structures

It'd be very cool if we could do principled verification of some pivotal tech.

- One option is to try to turn verifying some particular technology into a math problem somehow.
- Another option is to try to come up with a way to do principled verification beyond math.
- A (probably very hard) good toy problem here is to come up with a system in which physics olympiad problem solutions could be written and verified.

What property is a guardrail supposed to judge?

I'll be assuming that a guardrail/monitor/verifier is in the business of deciding if an AI's actions/plans/outputs are suspicious or fine.

- I don't really know what this should mean precisely.
- What the classifier is doing could depend on the context.
 - e.g., we could want a monitor to judge some thinking about how to convince humans suspicious if the agent was asked to come up with aligned ASI code, but fine if the agent was asked to write a book that gets ASI banned
 - e.g., we could have a verifier specifically trying to check that a proposed drug cures a particular disease
- It's imo weird to go with any $P(\text{harm})$ decision rule here, for a couple reasons:
 - A smart enough observer can probably see that any meaningful action causes quite a large quantity of particular harms in expectation.
 - Maybe it's better to go with something like $P(\text{this is better than some baseline})$.
 - Anyway, I don't think either really captures the right thing for e.g. small research actions.
- Anyway, I think we don't have to fix a precise choice for the purposes of this presentation.

Issues with making it so the agent does not know it is guardrailed/monitored

Here is a reason that'd be hard:

- If it's doing a long project, it basically needs to see its past actions.
- If its actions are being nontrivially guardrailed, it can see it has done some very surprising stuff in the past!
- It has a lot of data about what goes through. And it can run new experiments.
- So it's likely that by default it can fairly easily learn a lot about how it is being guardrailed.

If we're only verifying final outputs (like, we ask an AI for a technology, and then only try to verify the thing it proposes), then the above doesn't apply. Still, even in that case:

- The hypothesis "maybe I'm being guardrailed" is probably fairly obvious. It's hard to make the agent have a bad belief about this true fact of fairly obvious interest.